Statistics

Chapter 5 — Instructor Solutions Manual

October 9, 2026 • Total Exercises: 94

Chapter 5 **Exercise 1**

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PROBLEM STATEMENT

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Step 1

The following outcomes are possible: (favour, favour), (favour, against), (against, against), (against, favour) where the first value represent the first person s opinion and the second one represents second person s opinion.

PROBLEM STATEMENT

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Step 1

List of Possible Outcomes.

	Test 1	Test 2
1.	Acceptable	Acceptable
2.	Acceptable	Repairable
3.	Acceptable	Scrapped
4.	Repairable	Acceptable
5.	Repairable	Repairable
6.	Repairable	Scrapped
7.	Scrapped	Acceptable
8.	Scrapped	Repairable
9.	Scrapped	Scrapped

PROBLEM STATEMENT

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Solution A

Step 1

(a). To determine the probability that the selected student is a management major, remember that the total **number of students** surveyed was 34. Next, the number of **management major** surveyed was 6.

Step 2

With the number of management major and the total number of students surveyed, we simply **divide** 6 by 34 to obtain the probability P:

$$\begin{aligned} P &= \frac{6}{34} \\ &= 0.1765 \end{aligned}$$

With this, the **probability** that the selected student is a management major is 17.65%.

Step 3

(b). To know what concept of probability is best suited for this problem, first remember that **empirical probability** can be solved by dividing the number of times an event occurs by the total number of observations.

Step 4

From the problem, we can treat the student being a management major as an **event**. Thus, we can say that this event occurs 6 times. Then, the number of students surveyed is the number of observations of this problem. Thus, the total number of observations is 34.

Step 5

Given these, we use **empirical probability** to determine what are the chances that the event of selecting a management major will happen from a total of 34 observations.

Solution B

Step 1

Total number of students= 34

Number of students with a major in management= 6

a. Probability that a student is a management major

$$\begin{aligned} P(\text{a student is a management major}) &= \frac{\text{Number of students with a major in management}}{\text{Total number of students}} \\ &= \frac{6}{34} \\ &= 0.176 \end{aligned}$$

b. Empirical probability is used for estimating the above probability since the probability is calculated using the information available to us.

Result

a. 0.176

b. Empirical Probability

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PROBLEM STATEMENT

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Step 1

a. Number of candidates =5 and The number of students in a minority group=2

$$\begin{aligned}\text{Probability of an event} &= \frac{\text{Number of favorable events}}{\text{Total number of events}}\\ &= \frac{2}{5}\\ &= 0.4\end{aligned}$$

b. Classical Probability is used because each candidate is equally likely to get qualified.

Result

a. 0.4

b. Classical Probability

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PROBLEM STATEMENT

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Step 1

a. Empirical Probability because the probability that he gets a hit in his next bat is based on the similar events happened in the past.

b. Classical Probability because each of the 7 students is equally likely to become the chairperson.

c. Classical Probability because each of the 5 million tickets has an equal chance of winning the lottery.

d. Empirical Probability because the probability of 0.8 is based on the similar events happened in the past.

Chapter 5 Exercise 6

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PROBLEM STATEMENT

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Step 1

a. The chances of this experiment are :

- Both are man
- One man and one woman
- Both are woman

b. Classical Probability is used since each person is equally likely to be promoted.

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PROBLEM STATEMENT

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Step 1

a. The experiment is the survey of 40 oil industry executives.

b. One possible event is all executives answered "NO".

c.

$$\begin{aligned}\text{Probability of an event} &= \frac{\text{Number of times event occurs}}{\text{Total number of observations}} \\ &= \frac{10}{40} \\ &= 0.4\end{aligned}$$

d. Empirical Probability is used since the probability is calculated on the basis of similar events occurring in the past.

e. Each of the possible outcomes is not equally likely but all events are mutually exclusive since the response can be either yes or no. Both cannot happen together.

Chapter 5 Exercise 8

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PROBLEM STATEMENT

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Step 1

a. The experiment is reporting the number of violations of 2000 drivers.

b. One possible event is the a driver has 0 number of violations.

c.

Probability that a particular driver had exactly two speeding violations

$$\begin{aligned} &= \frac{\text{Number of times the event occurs}}{\text{Total number of observations}} \\ &= \frac{18}{2000} \\ &= 0.009 \end{aligned}$$

d. Empirical Probability because the probability of 0.009 is calculated on the basis of the similar events in the past.

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PROBLEM STATEMENT

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Step 1

$$a)\ Four\ possible\ outcomes\ are:$$
$$111,\ 123,\ 100,\ 321$$
$$b)\ Each\ of\ the\ digits\ of\ the\ PIN\ can\ be\ chosen\ in\ 10\ ways.$$
$$So,\ number\ of\ ways\ a\ PIN\ can\ be\ chosen = 10^3 = 1000$$
$$Probability = \frac{10^3 * 1}{10^3 * 10^3} = 0.001$$
$$c)\ Classical - Since\ any\ of\ the\ 10\ digits\ can\ be\ chosen\ with\ equal\ probability\ for\ each\ number\ o$$

PROBLEM STATEMENT

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Step 1

$$a)\ Some\ of\ the\ events\ are:$$
$$Prices\ of\ all\ shares\ increase$$
$$Price\ of\ only\ one\ of\ the\ shares\ increases,\ and\ that\ of\ remaining\ decrease.$$
$$b)\ For\ each\ share,\ the\ price\ can\ either\ increase\ with\ probability\ 0.5\ or\ decrease\ with\ probability$$
$$P(Prices\ of\ all\ shares\ increase) = 0.5^{100}$$
$$= 7.889 * 10^{-31}$$
$$P(Prices\ of\ only\ one\ of\ the\ shares\ increases) = \binom{100}{1} * 0.5^{1} * 0.5^{99} = 7.889 * 10^{-29}$$
$$c)\ Classical - Since\ the\ price\ of\ each\ share\ is\ equally\ likely\ to\ either\ increase\ or\ decrease.$$

PROBLEM STATEMENT

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Step 1

A and B are mutually exclusive events. This implies, $P(A \cap B) = 0$. $P(A) = 0.3$ and $P(B) = 0.20$.

So, Probability that either A or B will happen is

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= 0.30 + 0.20 - 0 \\ &= 0.50 \end{aligned}$$

Probability that neither A nor B will happen is

$$\begin{aligned} P(\sim A \cap \sim B) &= P(\sim (A \cup B)) \\ &= 1 - P(A \cup B) \\ &= 1 - 0.5 \\ &= 0.5 \end{aligned}$$

Result

$P(A \cap B) = 0.5$ and $P(\sim A \cap \sim B) = 0.5$

PROBLEM STATEMENT

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Step 1

X and Y are mutually exclusive events. This implies, $P(X \cap Y) = 0$. $P(X) = 0.05$ and $P(X) = 0.02$.

So, Probability that either X or Y will happen is

$$\begin{aligned} P(X \cup Y) &= P(X) + P(Y) - P(X \cap Y) \\ &= 0.05 + 0.02 - 0 \\ &= 0.07 \end{aligned}$$

Probability that neither X nor Y will happen is

$$\begin{aligned} P(\sim X \cap \sim Y) &= P(\sim (X \cup Y)) \\ &= 1 - P(X \cup Y) \\ &= 1 - 0.07 \\ &= 0.93 \end{aligned}$$

Result

$P(X \cup Y) = 0.07$ and $P(\sim X \cap \sim Y) = 0.93$

PROBLEM STATEMENT

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Solution A

Step 1

Recall : The formula of the probability of an event is given by;

$$\text{Probability of an event } = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$$

Step 2

Let A be the first event namely "Under $\$1$ million" with 102 number of firms, B be the second event namely "$\$1$ million to $\$20$ million" with 61 number of firms, and C be the third event namely "$\$20$ million or more" with 37 number of firms.

$(\mathbf{A.})$ To achieve the goal of this task, apply the formula of the probability on the first event A.

$$\begin{aligned} P(A) &= \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}} \\ &= \frac{102}{102+61+37} \\ &= \frac{102}{200} \\ &= 0.51 \text{ or } 51\%. \end{aligned}$$

Step 3

$(\mathbf{B.})$ To achieve the goal of the task, use the special rule of addition where;

$$P(A \text{ or } B) = P(A) + P(B)$$

Step 4

First, determine the probability of the two events.

For $P(B)$,

$$\begin{aligned} P(B) &= \frac{61}{102+61+37} \\ &= \frac{61}{200} \\ &= 0.305 \text{ or } 30.5\%. \end{aligned}$$

Step 5

For $P(C)$,

$$\begin{aligned} P(C) &= \frac{37}{102+61+37} \\ &= \frac{37}{200} \\ &= 0.185 \text{ or } 18.5\%. \end{aligned}$$

Step 6

Now, apply the special rule of addition.

$$\begin{aligned} P(B \text{ or } C) &= P(B) + P(C) \\ &= \frac{61}{200} + \frac{37}{200} \\ &= \frac{98}{200} \\ &= 0.49 \text{ or } 49\%. \end{aligned}$$

Solution B

Step 1

Given:

Income after Taxes	Number of firms
Under \$1 million	102
\$1 million to \$20 million	61
\$20 million or more	37
Total	200

(a) 102 of the 200 firms have under \$1 million in income after taxes.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{Under \$1 million}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{102}{200} = \frac{51}{100} = 0.51 = 51\%$$

Step 2

(b) 61 of the 200 firms have \$1 million to \$20 million in income after taxes, while 37 of the 200 firms have \$20 million or more in income after taxes.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{\$1 million to \$20 million}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{61}{200}$$
$$P(\text{\$20 million or more}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{37}{200}$$

Use the **Special rule of addition** for disjoint or mutually exclusive events: $P(A \cup B) = P(A \text{ or } B) = P(A) + P(B)$

$$\begin{aligned} &P(\text{\$1 million to \$20 million, or \$20 million or more}) \\ &= P(\text{\$1 million to \$20 million}) + P(\text{\$20 million or more}) \\ &= \frac{61}{200} + \frac{37}{200} \\ &= \frac{98}{200} \\ &= \frac{49}{100} \\ &= 0.49 \\ &= 49\% \end{aligned}$$

Result

(a) $\frac{51}{100} = 0.51 = 51\%$

(b) $\frac{49}{100} = 0.49 = 49\%$, Special rule of addition

Chapter 5

Exercise 14

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PROBLEM STATEMENT

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Step 1

a. Under addition rule, $P(\text{company will not lose money next quarter})$

$$\begin{aligned} &= P(\text{company will earn profit}) + P(\text{company will break even}) \\ &= 0.5 + 0.3 \\ &= 0.8 \end{aligned}$$

b. Under the complement rule,

$$\begin{aligned} P(\text{company will not lose money next quarter}) &= 1 - P(\text{company will lose money next quarter}) \\ &= 1 - 0.2 \\ &= 0.8 \end{aligned}$$

Result

a. 0.8

b. 0.8

PROBLEM STATEMENT

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Step 1

The grade will be above C if the grade is either A or B.

$P(A) = 0.25$ and $P(B) = 0.5$. Since, the student can't get the grade A and B simultaneously, A and B are mutually exclusive events. This implies $P(A \cap B) = 0$.

$$\begin{aligned} P(C) &= P(A \cup B) \\ &= P(A) + P(B) - P(A \cap B) \\ &= 0.25 + 0.5 \\ &= 0.75 \end{aligned}$$

Result

$$0.75$$

PROBLEM STATEMENT

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Step 1

If A is the event '2 heads' and B is the event '2 tails', then A and B are mutually exclusive because if A occurs, it is not possible for B to occur simultaneously. Similarly, if B occurs, A cannot occur. A and B are not complements, because if A occurs, it does not imply the occurrence of B. If the outcome is not 2 heads, it can be either '2 tails' or 'one head, one tail'. The sum of the probability of A and that of B is not 1.

🎯 Result

Yes, No

PROBLEM STATEMENT

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Solution A

Step 1

In this problem, we need to determine the probability of either A or B occurring using the given probabilities.

Step 2

The given events are A and B, and we are also given the probability of event A and B. To find the probability of A or B event, we will use the formula for the probability of two events that are not mutually exclusive given in the book:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Where P represents the probability of each event. Event A and B is represented with a standard denotation $A \cap B$ and the required event A or B is denoted as $A \cup B$.

Step 3

Probabilities given in the problem are:

Event	Probability
A	0.2
B	0.3
$A \cup B$	0.15

Step 4

Next, we can substitute the given data in the equation to calculate the required probability of event A or B.

$$P(A \cap B) = P(A) + P(B) - P(A \cup B)$$

$$= 0.2 + 0.3 - 0.15$$

$$= \boxed{0.35}$$

The probability of event A or B occurring is 0.35.

🎯 Result

$$P(A \cap B) = 0.35$$

Solution B

Step 1

P(A) = 0.2 P(B) = 0.3 P(A and B) = 0.15 P(Either A or B) = P(A) + P(B) - P(A and B) = 0.2 + 0.3 - 0.15 = 0.35

🎯 Result

0.35

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PROBLEM STATEMENT

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Step 1

$P(X) = 0.55$, $P(Y) = 0.35$ and $P(X \cap Y) = 0.2$.

$$\begin{aligned} P(X \cup Y) &= P(X) + P(Y) - P(X \cap Y) \\ &= 0.55 + 0.35 - 0.2 \\ &= 0.7 \end{aligned}$$

Result

$$0.7$$

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PROBLEM STATEMENT

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Step 1

If two events are mutually exclusive, by definition, their probability of joint occurrence will be zero. This is because when events A and B are mutually exclusive, then occurrence of one event implies the non-occurrence of the other.

Result

0

PROBLEM STATEMENT

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Step 1

Let H be the event that the student will pass the history course and M be the event that the student will pass the math course.

So, $P(H) = 0.60$ and $P(M) = 0.7$.

Probability of passing both the subjects is 0.5. This implies $P(H \cap M) = 0.50$.

The student passes at least one subject if he passes either H or M. So,

$$\begin{aligned} P(H \cup M) &= P(H) + P(M) - P(H \cap M) \\ &= 0.6 + 0.7 - 0.5 \\ &= 0.8 \end{aligned}$$

Result

$$0.8$$

PROBLEM STATEMENT

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Step 1

a) Since 20% of the stores have both a pharmacy and a floral shop, P (a randomly selected store has both pharmacy and floral shop) = 0.2

b) Since 30% of the stores have both pharmacy and deli, P (a randomly selected store has both pharmacy and deli) = 0.3

c) Since P(select a store with both pharmacy and deli) is non-zero, the events "select a store with deli" and "select a store with pharmacy" are not mutually exclusive.

d) It is a joint probability.

e) P (selecting a store that does not have all three departments) = 1 - 0.1 = 0.9

Result

a) 0.2 b) 0.3 c) No d) Joint probability e) 0.9

PROBLEM STATEMENT

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Step 1

Let A be the event that a vacationer visits Yellowstone Park and B be the event that a vacationer visits the Tetons.

P(A) = 0.5 P(B) = 0.4 P(A and B) = 0.35

a) P(A or B) = P(A) + P(B) - P(A and B) = 0.5 + 0.4 - 0.35 = 0.55

b) It is called joint probability.

c) The events are not mutually exclusive because P(A and B) is not zero.

Result

a) 0.55 b) Joint probability c) Not mutually exclusive

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PROBLEM STATEMENT

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Step 1

$P(A) = 0.4$ and $P(A|B) = 0.3$.

$$\begin{aligned} P(A \text{ and } B) &= P(A)P(B|A) \\ &= 0.4 \times 0.3 \\ &= 0.12 \end{aligned}$$

Result

$$0.12$$

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PROBLEM STATEMENT

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Step 1

$P(X_1) = 0.75$ and $P(Y_2|X_1) = 0.4$.

$$\begin{aligned} P(X_1 \text{ and } Y_2) &= P(X_1)P(Y_2|X_1) \\ &= 0.75 \times 0.4 \\ &= 0.3 \end{aligned}$$

Result

$$0.3$$

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PROBLEM STATEMENT

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Step 1

Let A be the event that the customer maintains a checking account and B be the event that the customer has a savings account.

$P(A) = 0.8$. $P(B) = 0.6$ and $P(A \text{ and } B) = 0.5$.

$$\begin{aligned} P(A \text{ or } B) &= P(A) + P(B) - P(A \text{ and } B) \\ &= 0.8 + 0.6 - 0.5 \\ &= 0.9 \end{aligned}$$

$$\begin{aligned} P(\text{Neither } A \text{ nor } B) &= 1 - P(A \text{ or } B) \\ &= 1 - 0.9 \\ &= 0.1 \end{aligned}$$

Result

$$P(A \text{ or } B) = 0.9 \text{ and } P(\text{Neither } A \text{ nor } B) = 0.1$$

PROBLEM STATEMENT

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Step 1

Let A be the event that first truck is available and B be the event that second truck is available.

$P(A) = 0.75$. $P(B) = 0.5$ and $P(A \text{ and } B) = 0.3$.

$$\begin{aligned} P(A \text{ or } B) &= P(A) + P(B) - P(A \text{ and } B) \\ &= 0.75 + 0.5 - 0.3 \\ &= 0.95 \end{aligned}$$

$$\begin{aligned} P(\text{Neither } A \text{ nor } B) &= 1 - P(A \text{ or } B) \\ &= 1 - 0.95 \\ &= 0.05 \end{aligned}$$

Result

$$0.05$$

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PROBLEM STATEMENT

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Step 1

$$a)\ P(A_1) = \frac{3}{10} = 0.3$$

$$b)\ P(B_1/A_2) = \frac{P(B_1\ and\ A_2)}{P(A_2)} = \frac{\frac{1}{10}}{\frac{3}{10}} = \frac{1}{3} = 0.333$$

$$c)\ P(B_2\ and\ A_3) = \frac{1}{10} = 0.1$$

Result

a) 0.3, b) 0.333, c) 0.1

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PROBLEM STATEMENT

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Step 1

a) P(first two toothbrushes will be defective) = $\frac{3}{20} * \frac{2}{19} = 0.0158$

b) P(first two toothbrushes will not be defective) = $\frac{17}{20} * \frac{16}{19} = 0.7158$

Result

a) 0.0158, b) 0.7158

PROBLEM STATEMENT

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Step 1

Given:

	Fair	Good	Excellent	Total
Below average	16	12	22	50
Average	45	60	45	150
Above average	93	72	135	300
Total	154	144	202	500

(a) The given table is called a two-way table or a **contingency table**.

Step 2

(b) The table contains information about 500 salespeople and thus there are 500 possible outcomes.

$$\#\text{ of possible outcomes} = 500$$

The value 135 is given in the row "Above average" and in the column "Excellent" of the given table, which implies that there are 135 favorable outcomes.

$$\#\text{ of favorable outcomes} = 135$$

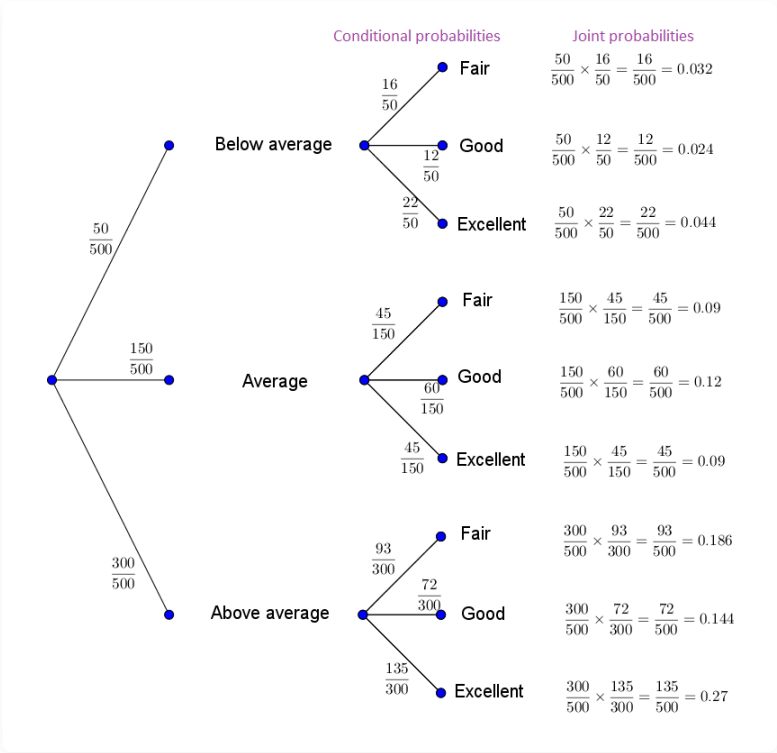
The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$\begin{aligned} P(\text{Above average and Excellent}) &= \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} \\ &= \frac{135}{500} \\ &= \frac{27}{100} \\ &= 0.27 \\ &= 27\% \end{aligned}$$

Step 3

(c) Let us first consider the variable Sales ability. Since there are 3 types of sales ability, the first level of the three diagram needs to have three vertices (which are all children of the roots). We label the edges by the proportion of salespersons with the particular sales ability.

Let us next consider the variable Potential for Advancement. Since there are 3 types of potential for advancement, each child at the first level of the tree needs to have three children labeled Fair, Good and Excellent. We label the edges by the proportion of salespersons with the particular type of potential for advancement among those with the particular sales ability.



Result

(a) Contingency table

(b) $\frac{27}{100} = 0.27 = 27\%$

(c) Tree diagram

PROBLEM STATEMENT

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Step 1

Let I be the event that a stock price increases, D be the event that it decreases, and R be the event that it remains the same. For each stock, P(I) = P(D) = P(R) = $\frac{1}{3}$.

For each stock, any of these three cases may happen. Therefore, total number of possible outcomes = $3 * 3 * 3 = 27$.

Some of the outcomes are: (I,I,I), (I,I,D), (I,D,I), (D,I,I), (I,R,R), (I,D,R) and so on.

P(at least two stocks increase in value) = P(only two stocks increase in value, and the other one either remains same or decreases) + P(all increase in value)

$=3 * \frac{1}{3} * \frac{1}{3} * \frac{2}{3} + (\frac{1}{3})^3 = \frac{2}{9} + \frac{1}{27} = 0.259$

Result

0.259

PROBLEM STATEMENT

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Step 1

DEFINITIONS

General multiplication rule:

$$P(A \text{ and } B) = P(A \cap B) = P(A) \times P(B|A) = P(B) \times P(A|B)$$

Step 2

SOLUTION

3 of the 5 people are strong leaders, which implies that the first salesperson has 3 chances out of 5 of being a strong leader and thus the first salesperson also have 3 chances out of 5 of winning the account.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{First salesperson wins}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{3}{5} = 0.6$$

The second salesperson could have only won when the first salesperson was not a strong leader and the second salesperson was a strong leader. The first person has 2 chances out of 5 of not being the strong leader, while the second person then has 3 chances remaining out of 4 remaining people of being the strong leader (when the first person was not a strong leader).

Use the general multiplication rule:

$$\begin{aligned}
&P(\text{Second salesperson wins}) \\
&= P(\text{First person is not strong and second person is strong}) \\
&= P(\text{First person is not strong}) \\
&\times P(\text{Second person is strong}|\text{First person is not strong}) \\
&= \frac{2}{5} \times \frac{3}{4} \\
&= \frac{6}{20} \\
&= \frac{3}{10} \\
&= 0.3
\end{aligned}$$

Step 3

The third salesperson could have only won when the first two salespersons were not strong leaders and the third salesperson was a strong leader. The first person has 2 chances out of 5 of not being the strong leader, while the second person then has 1 chance remaining out of 4 remaining people of not being a strong leader (when the first person was not a strong leader), while the third person then has 3 chances out of 3 of being a strong leader (when the first two people were not strong leaders).

Use the general multiplication rule:

$$
\begin{aligned}
&P(\text{Third salesperson wins}) \\
&= P(\text{First two people are not strong and second person is strong}) \\
&= P(\text{First person is not strong}) \\
&\times P(\text{Second person is not strong}|\text{First person is not strong}) \\
&\times P(\text{Third person is strong}|\text{First two people are not strong}) \\
&= \frac{2}{5} \times \frac{1}{4} \times \frac{3}{3} \\
&= \frac{6}{60} \\
&= \frac{1}{10} \\
&= 0.1
\end{aligned}
$$

We note that the probability that the third salesperson wins is much smaller than the probability that the first salesperson wins and thus the three salespeople do not have the same chance of winning the account.

🎯 Result

No, first salesperson has a 0.6 chance of winning, second salesperson has a 0.3 chance of winning, and third salesperson has a 0.1 chance of winning.

Chapter 5 Exercise 32

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PROBLEM STATEMENT

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Step 1

For each person, probability of being born on a Wednesday = $\frac{1}{7}$

a) P(all three were born on Wednesday) = $(\frac{1}{7})^3 = \frac{1}{343} = 0.0029$

b) P(all were born on different days of the week) = $\frac{7}{7} * \frac{6}{7} * \frac{5}{7} = \frac{30}{49} = 0.6122$

c) P(None of them was born on Saturday) = $\frac{6}{7} * \frac{6}{7} * \frac{6}{7} = \frac{216}{343} = 0.6297$

Chapter 5 Exercise 33

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PROBLEM STATEMENT

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Step 1

According to Bayes' Theorem,

$$P(A_1/B_1) = \frac{P(B_1/A_1) * P(A_1)}{P(B_1/A_1) * P(A_1) + P(B_1/A_2) * P(A_2)} = \frac{0.05 * 0.6}{0.05 * 0.6 + 0.1 * 0.4} = 0.4286$$

Result

0.4286

PROBLEM STATEMENT

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Step 1

According to Bayes' Theorem,

$$P(A_3/B_1) = \frac{P(B_1/A_3) * P(A_3)}{P(B_1/A_1) * P(A_1) + P(B_1/A_2) * P(A_2) + P(B_1/A_3) * P(A_3)} = \frac{0.1 * 0.4}{0.25 * 0.2 + 0.05 * 0.}$$

Result

0.3636

PROBLEM STATEMENT

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Step 1

P(night game) = 0.7

P(day game) = 0.3

P(winning / night game) = 0.5

P(winning / day game) = 0.9

Use Bayes' theorem:

P(night game / winning) =

$$\frac{P(winning/night\ game) * P(night\ game)}{P(winning/night\ game) * P(night\ game) + P(winning/day\ game) * P(day\ game)} = \frac{0.5 * 0.7}{0.5 * 0.7 + 0.9 * 0.3} = \frac{0.35}{0.62} = \frac{35}{62} \approx 0.5645$$

Result

0.5645

PROBLEM STATEMENT

Chapter 5, Page 171

Step 1

Let A_1 be the event that the students will complete the assignment and A_2 be the event that the students will not complete the assignment. Let B be the event that the student will pass.

$P(A_1) = 0.8$, $P(A_2) = 0.2$, $P(B|A_1) = 0.90$ and $P(B|A_2) = 0.6$. Now,

$$\begin{aligned} P(A_1|B) &= \frac{P(A_1)P(B|A_1)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2)} \\ &= \frac{0.8 \times 0.90}{0.8 \times 0.9 + 0.2 \times 0.6} \\ &= \frac{0.72}{0.72 + 0.12} \\ &= \frac{0.72}{0.84} \\ &= 0.8571 \end{aligned}$$

Result

$$0.8571$$

PROBLEM STATEMENT

Chapter 5, Page 171

Step 1

Let A_1 be the event that the sales are in cash or check, A_2 be the event that sales are paid with the credit card and A_3 be the event that the sales are paid with the debit card. Let B be the event that the purchases are for more than $\$50$.

$P(A_1) = 0.3$, $P(A_2) = 0.3$, $P(A_3) = 0.4$, $P(B|A_1) = 0.20$, $P(B|A_2) = 0.9$ and $P(B|A_3) = 0.6$.

Now,

$$\begin{aligned} P(A_1|B) &= \frac{P(A_1)P(B|A_1)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)} \\ &= \frac{0.3 \times 0.20}{0.3 \times 0.20 + 0.3 \times 0.9 + 0.4 \times 0.6} \\ &= \frac{0.06}{0.06 + 0.27 + 0.24} \\ &= \frac{0.06}{0.57} \\ &= 0.1053 \end{aligned}$$

Result

$$0.1053$$

PROBLEM STATEMENT

Chapter 5, Page 171

Step 1

Let A_1 be the event that the garage doors are open and A_2 be the event that the garage doors are not open. Let B be the event that something will be stolen.

$P(A_1) = \frac{1}{4} = 0.25$, $P(A_2) = \frac{3}{4} = 0.75$, $P(B|A_1) = 0.05$ and $P(B|A_2) = 0.01$. Now,

$$\begin{aligned} P(A_1|B) &= \frac{P(A_1)P(B|A_1)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2)} \\ &= \frac{0.25 \times 0.05}{0.25 \times 0.05 + 0.75 \times 0.01} \\ &= \frac{0.0125}{0.0125 + 0.0075} \\ &= \frac{0.0125}{0.02} \\ &= 0.625 \end{aligned}$$

Result

$$0.625$$

PROBLEM STATEMENT

Chapter 5, Page 176

Solution A

Step 1

Recall : The notation $n!$ is called n factorial which means the product of $n(n-1)(n-2)(n-3)\ldots(1)$.

$(\mathbf{A.})$ Apply the definition of the n factorial to achieve the goal of the task.

$$\begin{aligned}\frac{n!}{n!} &= \frac{n(n-1)(n-2)(n-3)\ldots(1)}{n(n-1)(n-2)(n-3)\ldots(1)} \\ &= \frac{40\cdot 39\cdot 38\cdot 37\cdot 36\cdot \cancel{35!}}{\cancel{35!}} \qquad \text{Simply cancel out common factors.} \\ &= \frac{40\cdot 39\cdot 38\cdot 37\cdot 36}{1} \\ &= \boxed{78960960}\end{aligned}$$

Step 2

$(\mathbf{B.})$ Recall : The permutation formula is given by

$$ {}_nP_r = \frac{n!}{(n-r)!}$$

Now, use this formula to achieve this goal of the task.

$$\begin{aligned} {}_nP_r &= \frac{n!}{(n-r)!} \\ &= \frac{7!}{(7-4)!} \\ &= \frac{7!}{3!} \\ &= \frac{7\cdot 6\cdot 5\cdot 4\cdot \cancel{3!}}{\cancel{3!}} \qquad \text{Cancel common factors.} \\ &= 7\cdot 6\cdot 5\cdot 4 \\ &= \boxed{840}\end{aligned}$$

Step 3

$(\mathbf{C.})$ Recall : the combination formula is given by

$$ {}_nC_r = \frac{n!}{r!(n-r)!} $$

Now, use this formula to achieve the goal of this task.

$$
\begin{aligned}
{}_nC_r &= \frac{n!}{r!(n-r)!} \\
&= \frac{5!}{2!(5-2)!} \\
&= \frac{5!}{2!3!} \\
&= \frac{5 \cdot 4 \cdot \cancel{3!}}{2 \cdot 1 \cdot \cancel{3!}} \qquad \text{Cancel common factors.} \\
&= \frac{5 \cdot 4}{2 \cdot 1} \\
&= \frac{20}{2} \\
&= \boxed{10}
\end{aligned}
$$

Solution B

Step 1

a.

$$\begin{aligned}\frac{40!}{35!} &= \frac{40 \times 39 \times 38 \times 37 \times 36 \times 35 \times 34 \times \ldots}{35 \times 34 \times ..} \\ &= 40 \times 39 \times 38 \times 37 \times 36 \\ &= 78960960\end{aligned}$$

b.

$$^{n}P_r = \frac{n!}{(n-r)!}$$

$$\begin{aligned}{}^{7}P_4 &= \frac{7!}{(7-4)!} \\ &= \frac{7!}{3!} \\ &= \frac{7 \times 6 \times 5 \times 4 \times 3!}{3!} \\ &= 840\end{aligned}$$

c.

$$^{n}C_r = \frac{n!}{r!(n-r)!}$$

$$\begin{aligned}{}^{5}C_2 &= \frac{5!}{2!(5-2)!} \\ &= \frac{5!}{2!3!} \\ &= \frac{5 \times 4 \times 3!}{2 \times 3!} \\ &= 10\end{aligned}$$

Result

a. 78960960

b. 840

c. 10

PROBLEM STATEMENT

Chapter 5, Page 176

Step 1

a) $\dfrac{20!}{17!} = 20 * 19 * 18 = 6840$

b) ${}^{9}P_{3} = \dfrac{9!}{(9-3)!} = 9 * 8 * 7 = 504$

c) ${}^{7}C_{2} = \dfrac{7!}{2! * (7-2)!} = \dfrac{7 * 6}{2 * 1} = 21$

PROBLEM STATEMENT

Chapter 5, Page 176

Step 1

DEFINITIONS

Multiplication formula: If the first event could occur in m ways and the second event could occur in n ways, then the number of ways that the two events could occur in sequence is $m \cdot n$.

Definition permutation (order is important):

$$ {}_nP_r = \frac{n!}{(n-r)!} $$

Definition combination (order is not important):

$$ {}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} $$

with $n! = n \cdot (n-1) \cdot \ldots \cdot 2 \cdot 1$.

Step 2

SOLUTION

We are selecting $r = 4$ of the $n = 10$ people.

$$n = 10$$
$$r = 4$$

The order in which the people are selected is not important, as a different order of the selected people results in the same group. This then implies that we need to use the definition of a **combination**.

$$\begin{aligned} {}_{10}C_4 &= \frac{10!}{4!(10-4)!} \\ &= \frac{10!}{4!6!} \\ &= \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{4 \cdot 3 \cdot 2 \cdot 1 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} \\ &= \frac{10 \cdot 9 \cdot 8 \cdot 7}{4 \cdot 3 \cdot 2 \cdot 1} \\ &= \frac{5040}{24} \\ &= 210 \end{aligned}$$

Thus there are 210 different groups of 4 possible.

Result

210 different groups

PROBLEM STATEMENT

Chapter 5, Page 176

Step 1

DEFINITIONS

Multiplication formula: If the first event could occur in m ways and the second event could occur in n ways, then the number of ways that the two events could occur in sequence is $m \cdot n$.

Definition permutation (order is important):

$$ {}_nP_r = \frac{n!}{(n-r)!} $$

Definition combination (order is not important):

$$ {}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} $$

with $n! = n \cdot (n-1) \cdot \ldots \cdot 2 \cdot 1$.

Step 2

SOLUTION

We are interested in phone numbers of 7 digits that start with 537 and thus we need to determine the number of ways to select the last 4 digits of the telephone number. Since there are 10 possible values for a digit (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, and 10), each of the four digits has 10 possible outcomes.

First digit: 10 ways

Second digit: 10 ways

Third digit: 10 ways

Fourth digit: 10 ways

Use the multiplication formula:

$$ \#\text{ of telephone numbers} = 10 \cdot 10 \cdot 10 \cdot 10 = 10,000 $$

Thus there are 10,000 different telephone numbers possible within the 537 exchange.

🎯 **Result**

10,000 different telephone numbers

PROBLEM STATEMENT

Chapter 5, Page 176

Solution A

Step 1

DEFINITIONS

Multiplication formula: If the first event could occur in m ways and the second event could occur in n ways, then the number of ways that the two events could occur in sequence is $m \cdot n$.

Definition permutation (order is important):

$$ {}_nP_r = \frac{n!}{(n-r)!} $$

Definition combination (order is not important):

$$ {}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} $$

with $n! = n \cdot (n-1) \cdot \ldots \cdot 2 \cdot 1$.

Step 2

SOLUTION

We are selecting $r = 5$ of the $n = 5$ cities.

$$n = 5$$
$$r = 5$$

The order in which the cities are selected is important, as a different order of the selected cities results in a different route. This then implies that we need to use the definition of a **permutation**.

$$\begin{aligned} {}_5P_5 &= \frac{5!}{(5-5)!} \\ &= \frac{5!}{0!} \\ 0! &= 1 \\ &= \frac{5!}{1} \\ &= 5! \\ &= 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \\ &= 120 \end{aligned}$$

Thus there are 120 different possible routes.

Result

120 routes

Solution B

Step 1

The first city can be selected in 5 ways, second in 4 ways, and so on.

Therefore, number of routes possible = $5 * 4 * 3 * 2 * 1 = 120$

Result

120

PROBLEM STATEMENT

Chapter 5, Page 176

Step 1

DEFINITIONS

Multiplication formula: If the first event could occur in m ways and the second event could occur in n ways, then the number of ways that the two events could occur in sequence is $m \cdot n$.

Definition permutation (order is important):

$$ {}_nP_r = \frac{n!}{(n-r)!} $$

Definition combination (order is not important):

$$ {}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} $$

with $n! = n \cdot (n-1) \cdot \ldots \cdot 2 \cdot 1$.

Step 2

SOLUTION

We are selecting $r = 10$ of the $n = 15$ landfills.

$$n = 15$$
$$r = 10$$

The order in which the landfills are selected is not important, as a different order of the selected landfills results in the same sample. This then implies that we need to use the definition of a **combination**.

$$\begin{aligned} {}_{15}C_{10} &= \frac{15!}{10!(15-10)!} \\ &= \frac{15!}{10!5!} \\ &= \frac{15 \cdot 14 \cdot 13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} \\ &= \frac{15 \cdot 14 \cdot 13 \cdot 12 \cdot 11}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} \\ &= \frac{360,360}{120} \\ &= 3003 \end{aligned}$$

Thus there are 3003 possible different samples.

Result

3003 samples

PROBLEM STATEMENT

Chapter 5, Page 176

Solution A

Step 1

DEFINITIONS

Multiplication formula: If the first event could occur in m ways and the second event could occur in n ways, then the number of ways that the two events could occur in sequence is $m \cdot n$.

Definition permutation (order is important):

$$ {}_nP_r = \frac{n!}{(n-r)!} $$

Definition combination (order is not important):

$$ {}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} $$

with $n! = n \cdot (n-1) \cdot \ldots \cdot 2 \cdot 1$.

Step 2

SOLUTION

We are selecting $r = 10$ of the $n = 15$ questions.

$$n = 15$$
$$r = 10$$

The order in which the questions are selected is important, as a different order of the selected questions results in a different arrangement. This then implies that we need to use the definition of a **permutation**.

$$\begin{aligned} {}_{15}P_{10} &= \frac{15!}{(15-10)!} \\ &= \frac{15!}{5!} \\ &= \frac{15 \cdot 14 \cdot 13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} \\ &= 15 \cdot 14 \cdot 13 \cdot 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \\ &= 10,897,286,400 \end{aligned}$$

Thus there are 10,897,286,400 different arrangements.

Result

10,897,286,400 arrangements

Solution B

Step 1

Number of different arrangements for the order of 10 selected questions is

$$\begin{aligned} {}^{15}P_1 &= \frac{n!}{(n-r)!} \\ &= \frac{15!}{(15-10)!} \\ &= \frac{15!}{5!} \\ &= 15 \times 14 \times 13 \times .....7 \times 6 \\ &= 10897286400 \end{aligned}$$

🎯 Result

10897286400

PROBLEM STATEMENT

Chapter 5, Page 176

Step 1

DEFINITIONS

Multiplication formula: If the first event could occur in m ways and the second event could occur in n ways, then the number of ways that the two events could occur in sequence is $m \cdot n$.

Definition permutation (order is important):

$$ {}_nP_r = \frac{n!}{(n-r)!} $$

Definition combination (order is not important):

$$ {}_nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!} $$

with $n! = n \cdot (n-1) \cdot \ldots \cdot 2 \cdot 1$.

Step 2

SOLUTION

We are selecting $r = 3$ of the $n = 7$ managers as division heads.

$$n = 7$$
$$r = 3$$

The order in which the managers are selected is important, as a different order of the selected managers results in a different assignment (as the division assignment makes a difference). This then implies that we need to use the definition of a **permutation**.

$$\begin{aligned} {}_7P_3 &= \frac{7!}{(7-3)!} \\ &= \frac{7!}{4!} \\ &= \frac{7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{4 \cdot 3 \cdot 2 \cdot 1} \\ &= 7 \cdot 6 \cdot 5 \\ &= 210 \end{aligned}$$

Thus there are 210 different ways that the three new heads can be appointed.

🎯 Result

210 ways

Chapter 5 **Exercise 47** Page 178

PROBLEM STATEMENT

Chapter 5, Page 178

Step 1

a) The experiment involves asking teenagers to compare a newly developed soft drink with their favorite soft drink.

b) One possible event is that all the teenagers surveyed liked it better than their favorite soft drink.

Chapter 5 Exercise 48

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PROBLEM STATEMENT

Chapter 5, Page 178

Step 1

This approach is called Empirical probability. Empirical probability calculates the probability of an event as the fraction of the times a similar event happened in the past

Result

Empirical probability

Chapter 5 Exercise 49

Page 178

PROBLEM STATEMENT

Chapter 5, Page 178

Step 1

This illustrates subjective probability, since this probability has been arrived at subjectively by an individual using the available information.

Result

Subjective Probability

PROBLEM STATEMENT

Chapter 5, Page 178

Step 1

a. Probability of an event$= \dfrac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$.

$$\begin{aligned} \Rightarrow P(\text{tip of more than \$200 or more}) &= \frac{50}{500} \\ &= 0.1 \end{aligned}$$

b. All the categories are mutually exclusive events since the amount of tip earned cannot lie in the two categories at the same time.

c. The sum of the probabilities of each outcome is 1 because all the outcomes are mutually exclusive and exhaustive. The amount of tip cannot be outside these categories.

d.

$$\begin{aligned} P(\text{tip of upto \$50}) &= \frac{\text{number of checks earning tip of 0 upto \$20+tip of \$20 upto \$50}}{\text{Total sample}} \\ &= \frac{200+100}{500} \\ &= 0.6 \end{aligned}$$

e.

$$\begin{aligned} P(\text{tip less than \$200} &= 1 - P(\text{tip \$200 or more}) \\ &= 1 - 0.1 \ \text{ from } \ (\text{a.}) \\ &= 0.9 \end{aligned}$$

🎯 Result

a. 0.1

b. Mutually exclusive events

c. 1

d. 0.6

e. 0.9

Chapter 5 **Exercise 51** Page 178

PROBLEM STATEMENT

Chapter 5, Page 178

Step 1

P(wins Preakness Stakes) = $\frac{2}{3}$

a) P(wins Belmont Stakes) = $\frac{2}{3}$

P(wins the Triple Crown) = $\frac{2}{3} * \frac{2}{3} = \frac{4}{9} = 0.444$

b) Let the probability of winning the Preakness Stakes be x.

To be even money, P(wins the Triple Crown) = 0.5

$$0.5 = \frac{2}{3} * x$$
$$x = \frac{3}{4} = 0.75$$

Result

$$a)\ \frac{4}{9},\ b)\ 0.75$$

PROBLEM STATEMENT

Chapter 5, Page 178

Step 1

a) If the first king is returned to the deck, then on the second selection, again there are four king cards in a total of 52.

So, P(king is drawn) = $\frac{4}{52} = \frac{1}{13} = 0.0769$

b) If the king is not replaced, there are only 3 king cards available out of a total of 51 cards in the second selection.

P(king is drawn) = $\frac{3}{51} = 0.0588$

c) P(king is drawn on both the draws) = $\frac{4}{52} * \frac{3}{51} = \frac{1}{221} = 0.00452$

Result

a) $\frac{1}{13} = 0.0769$

b) $\frac{3}{51} = 0.0588$

c) $\frac{1}{221} = 0.00452$

PROBLEM STATEMENT

Chapter 5, Page 178

Step 1

P(system lasts 3 years) = 0.95

a) Since all the systems operate independently of each other,

P(all 4 systems last 3 years) = $0.95^4 = 0.8145$

b) This illustrates the special rule of multiplication, since each system is independent of others.

c) Let the event that the first, second, third, and fourth system operate properly for at least 3 years be represented by A, B, C, D respectively.

Then, $P(A \text{ and } B \text{ and } C \text{ and } D) = P(A) * P(B) * P(C) * P(D)$

Result

a) 0.8145

b) Special rule of multiplication

c) $P(A \text{ and } B \text{ and } C \text{ and } D) = P(A) * P(B) * P(C) * P(D)$

PROBLEM STATEMENT

Chapter 5, Page 178

Step 1

a) It is a Venn diagram.

b) It represents the complement rule, since it shows that the probability of an event occurring and that of the event not occurring sum to one.

c) B represents the event of choosing a family that receives welfare payments.

$P(B) + P(\sim B) = 1$ according to the complement rule.

$P(B) + P(\sim B)$ represents the event of choosing a family that receives welfare payments or a family that does not receive welfare payments.

Result

a) Venn diagram

b) Complement rule

c) 1

PROBLEM STATEMENT

Chapter 5, Page 179

Step 1

Let F represent female, $\sim F$ represent male, and C represent attend college.

$$P(F) = 80\% = 0.80$$
$$P(\sim F) = 20\% = 0.20$$
$$P(C|F) = 90\% = 0.90$$
$$P(C|\sim F) = 78\% = 0.78$$

(a) When 90% of the females attended college, then $100\% - 90\% = 10\%$ of the females do not attend college.

$$P(\sim C|F) = 10\% = 0.10$$

Use the **General multiplication rule**: $P(A \cap B) = P(A) \times P(B|A) = P(B) \times P(A|B)$

$$\begin{aligned} &P(\sim C \text{ and } F) \\ &= P(F) \times P(\sim C|F) \\ &= 0.80 \times 0.10 \\ &= 0.08 \\ &= 8\% \end{aligned}$$

Step 2

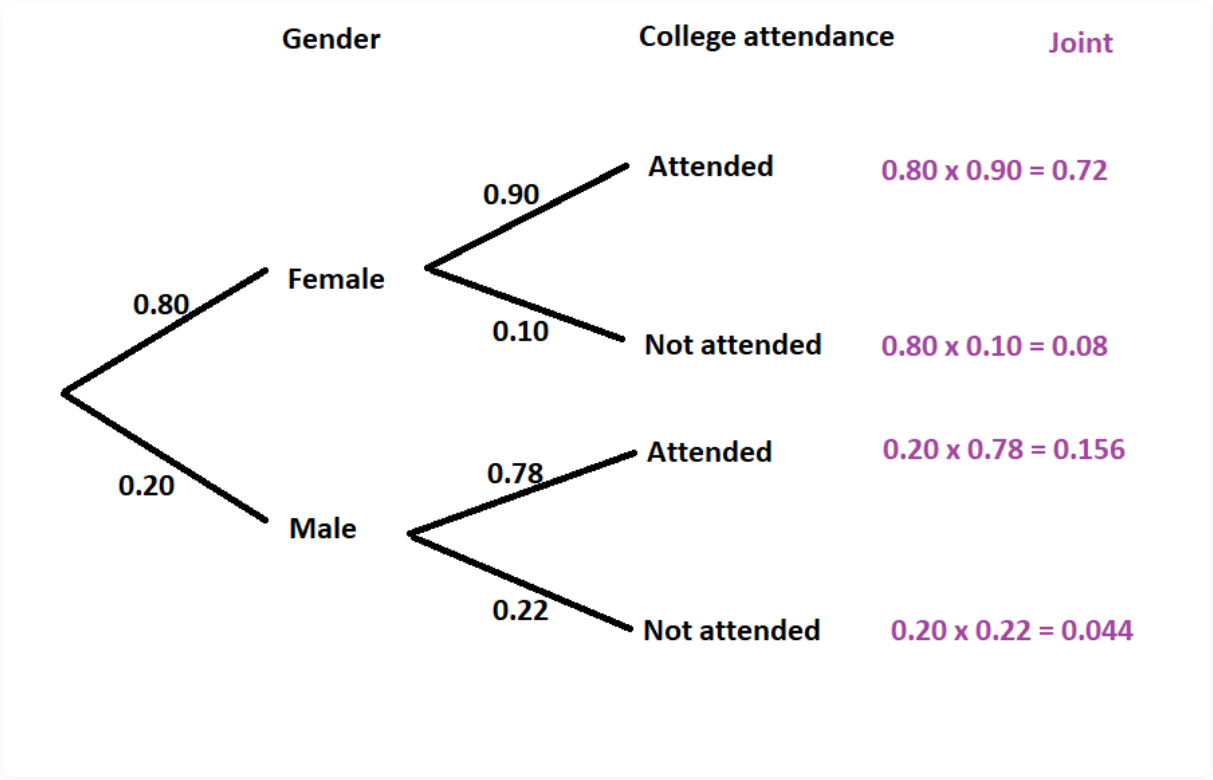
(b) Two events are **independent**, if the probability that one event occurs in no way affects the probability of the other event occurring.

When gender and attending college are independent, then the same proportion of males and females attend college. However, 90% of the females attend college and 78% of the males attend college. Since these percentages are not the same, gender and attending college are $not\ independent$.

Step 3

(c) Let the first level of the tree diagram contain the genders. This then implies that there are two vertices at the first level of the tree: Female and Male. 80% of the people are female, while 20% of the people are male.

Let the second level of the tree diagram represent college attendance. This then implies that there are two children (Attended and not attended) at the second level for each vertex at level 1. 90% of the females attend college, thus $100\% - 90\% = 10\%$ of the females do not attend college. 78% of the males attend college, thus $100\% - 78\% = 22\%$ of the males do not attend college.



Step 4

(d) Let us add all joint probabilities in the tree diagram of part (c):

$$0.72 + 0.08 + 0.156 + 0.044 = 1.000$$

We note that the joint probabilities do total 1.00, because the tree diagram contains all possible outcomes.

Result

(a) $0.08 = 8\%$

(b) Not independent

(c) See tree diagram

(d) Yes

Chapter 5

Exercise 56

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PROBLEM STATEMENT

Chapter 5, Page 179

Step 1

Let A be the event that a flight arrives within 15 minutes of the scheduled time.

P(A) = 0.9

a) Since all flights are independent of each other,

P(all 4 flights arrived within 15 minutes of the scheduled time) = $0.9^4 = 0.6561$

b) P(none of the flights arrived within 15 minutes of the scheduled time) = $0.1^4 = 0.0001$

c) P(at least one flight did not arrive within 15 minutes of the scheduled time) =

1 - P(all flights arrived within 15 minutes of the scheduled time)

$=1 - 0.6561 = 0.3439$

Result

a) 0.6561

b) 0.0001

c) 0.3439

PROBLEM STATEMENT

Chapter 5, Page 179

Step 1

P(production worker) = 0.57 P(supervisor) = 0.4 P(secretary) = 0.02 P(President) = 0.01

a) P(a randomly selected employee is a production worker) = 0.57 b) P(Either a production worker or supervisor) = P(production worker) + P(supervisor) = 0.57 + 0.4 = 0.97 c) The events are mutually exclusive since: P(A person is both a production worker and a supervisor) = 0 (from the description given in question) d) P(Neither a production worker nor a supervisor) = 1 - P((Either a production worker or supervisor) = 1 - 0.97 = 0.03

Result

a) 0.57, b) 0.97, c) Yes, d) 0.03

PROBLEM STATEMENT

Chapter 5, Page 179

Step 1

a) This is an example of empirical probability since here, probability is the fraction of times similar events have happened.

b) P(three hits in the game) = P(Getting a hit for all three times he batted) = $0.365^3 = 0.0486$

c) P(not getting any hit) = P(not getting a hit for any time he batted) = $(1 - 0.365)^3 = 0.635^3 = 0.2560$

d) P(getting at least one hit) = 1 - P(not getting any hit) = $1 - 0.2560 = 0.744$

Result

a) Empirical probability, b) 0.0486, c) 0.2560, d) 0.744

PROBLEM STATEMENT

Chapter 5, Page 179

Step 1

P(team favored in the semi finals wins it) = $\frac{2}{3}$

P(the other favored team wins its game) = $\frac{3}{4}$

a) P(Both favored teams win their games) = $\frac{2}{3} * \frac{3}{4} = \frac{1}{2} = 0.5$

b) P(Neither favored team wins its game) = $(1 - \frac{2}{3}) * (1 - \frac{3}{4}) = \frac{1}{12}$

c) P(at least one of the favored teams wins) = 1 - P(neither favored team wins its game) = $1 - \frac{1}{12} = \frac{11}{12}$

Result

a) 0.5

b) $\frac{1}{12}$

c) $\frac{11}{12}$

PROBLEM STATEMENT

Chapter 5, Page 179

Step 1

DEFINITIONS

Multiplication rule for independent events:

$$P(A \cap B) = P(A \text{ and } B) = P(A) \times P(B)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A \cup B) = P(A \text{ or } B) = P(A) + P(B)$$

Step 2

SOLUTION

(a) The three contenders are equally matched, which implies that each of the contestant is equally likely to find one of the "daily doubles. Moreover, this then implies that each contender has 1 chance in 3 of finding the first daily double.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$\begin{aligned} P(\text{Contestant finds daily double}) &= \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} \\ &= \frac{1}{3} \end{aligned}$$

Assuming that the selection of a "daily double" does not depend on the selection of the other "daily doubles", it is appropriate to use the multiplication rule for independent events:

$$\begin{aligned} &P(\text{Contestant finds all daily doubles}) \\ &= P(\text{Contestant finds first daily double}) \\ &\times P(\text{Contestant finds second daily double}) \\ &\times P(\text{Contestant finds third daily double}) \\ &= (P(\text{Contestant finds daily double}))^3 \\ &= \left(\frac{1}{3}\right)^3 \\ &= \frac{1}{27} \end{aligned}$$

Step 3

However, there are three contestants and all three of the contestants have a $\frac{1}{27}$ chance of finding all three daily doubles. Since two contestants cannot both find all three daily doubles, the events are mutually exclusive and thus it is appropriate to use the addition rule for mutually exclusive events:

$$
\begin{aligned}
&P(\text{Some contestant finds all daily doubles}) \\
&= P(\text{First contestant finds all daily doubles}) \\
&+ P(\text{Second contestant finds all daily doubles}) \\
&+ P(\text{Third contestant finds all daily doubles}) \\
&= 3P(\text{Contestant finds all daily doubles}) \\
&= 3 \times \frac{1}{27} \\
&= \frac{1}{9} \\
&\approx 0.1111 \\
&= 11.11\%
\end{aligned}
$$

Step 4

(b) The three contenders are equally matched, which implies that each of the contestant is equally likely to find one of the "daily doubles. Moreover, this then implies that each contender has 1 chance in 3 of finding the first daily double.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$\begin{aligned} P(\text{Contestant finds daily double}) &= \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} \\ &= \frac{1}{3} \end{aligned}$$

Assuming that the selection of a "daily double" does not depend on the selection of the other "daily doubles", it is appropriate to use the multiplication rule for independent events:

$$\begin{aligned} &P(\text{Returning champion all daily doubles}) \\ &= P(\text{Contestant finds first daily double}) \\ &\times P(\text{Contestant finds second daily double}) \\ &\times P(\text{Contestant finds third daily double}) \\ &= (P(\text{Contestant finds daily double}))^3 \\ &= \left(\frac{1}{3}\right)^3 \\ &= \frac{1}{27} \\ &\approx 0.0370 \\ &= 3.70\% \end{aligned}$$

Step 5

(c) Some contestant will always find the first daily double, while the probability of a certain event is always equal to 1.

$$P(\text{Some contestant finds first daily double}) = 1$$

After the first daily double has been selected by some contestant, there are 2 out of 3 contestants remaining that can select the second daily double (such that no contestant selects more than two daily doubles).

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$\begin{aligned} &P(\text{Other contestant finds second daily double}|\text{Some contestant first}) \\ &= \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} \\ &= \frac{2}{3} \end{aligned}$$

After the first two daily doubles have been selected by some contestant, there are 1 out of 3 contestants remaining that can select the third daily double (such that no contestant selects more than two daily doubles).

$$\begin{aligned} &P(\text{Other contestant finds third daily double}|\text{Two contestants found first two}) \\ &= \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} \\ &= \frac{1}{3} \end{aligned}$$

Step 6

Assuming that the selection of a "daily double" does not depend on the selection of the other "daily doubles", it is appropriate to use the multiplication rule for independent events:

$$\begin{aligned} &P(\text{Each contestant finds one daily double}) \\ &= P(\text{Some contestant finds first daily double}) \\ &\times P(\text{Other contestant finds second daily double}|\text{Some contestant first}) \\ &\times P(\text{Other contestant finds third daily double}|\text{Two contestants found first two}) \\ &= 1 \times \frac{2}{3} \times \frac{1}{3} \\ &= \frac{2}{9} \\ &\approx 0.2222 \\ &= 22.22\% \end{aligned}$$

Result

(a) $\frac{1}{9} \approx 0.1111 = 11.11\%$

(b) $\frac{1}{27} \approx 0.0370 = 3.70\%$

(c) $\frac{2}{9} \approx 0.2222 = 22.22\%$

Chapter 5 **Exercise 61**

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PROBLEM STATEMENT

Chapter 5, Page 179

Step 1

P(a 60 year old man survives another year) = 0.98

a) P(All five men survive the year) = $0.98^5 = 0.9039$

b) P(At least one does not survive) =

1 - P(all five men survive) = $1 - 0.9039 = 0.0961$

Result

a) 0.9039, b) 0.0961

PROBLEM STATEMENT

Chapter 5, Page 179

Step 1

P(a home includes a security system) = 0.4

a) P(all 3 homes have a security system) = $0.4^3 = 0.064$

b) P(none of the 3 homes has a security system) = $(1 - 0.4)^3 = 0.216$

c) P(at least one of the 3 homes has a security system) =

1 - P(none of the 3 homes has a security system) = $1 - 0.216 = 0.784$

d) Events were assumed to be independent, since the probability of selecting a particular kind of house is independent of previously selected houses.

Result

a) 0.064, b) 0.216, c) 0.784, d) Independent

PROBLEM STATEMENT

Chapter 5, Page 179

Step 1

a) P(all three homes have a security system) = P(selecting any three homes out of the four homes having security system) = $(4/10) * (3/9) * (2/8) = 0.0333$

b) P(none of the three homes has a security system) = P(selecting any three out of the six homes not having a security system) = $(6/10) * (5/9) * (4/8) = 0.1667$

c) P(at least one of the three homes has a security system) = 1 - P(none of the homes has a security system) = $1 - 0.1667 = 0.8333$

d) Events were assumed to be dependent since probability of selecting a particular house depends on the previously selected houses.

Result

a) 0.0333, b) 0.1667, c) 0.8333, d) Dependent

Chapter 5 Exercise 64

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PROBLEM STATEMENT

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Step 1

Given: Out of 20 families, 10 prepared their own taxes, 7 got it prepared by a local professional and 3 by H&R Block.

a) P(Selecting a family that prepared their own taxes) = $10/20 = 0.5$

b) P(Selecting two families, both of which prepared their own taxes) = $(10/20) * (9/19) = 0.2368$

c) P(Selecting three families, all of which prepared their own taxes) = $(10/20) * (9/19) * (8/18) = 0.1052$

d) P(Selecting two families, neither of which had their taxes prepared by H&R Block) = $(17/20) * (16/19) = 0.7158$

Chapter 5 Exercise 65

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PROBLEM STATEMENT

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Step 1

Given: Out of 12 members, 3 are women.

a) P(all three members are men) = $\binom{9}{3}/\binom{12}{3} = (9 * 8 * 7)/(12 * 11 * 10) = 0.3818$

b) P(at least one member is a woman) = 1 - P(all members are men) = $1 - 0.38181 = 0.61818$

Result

a) 0.3818, b) 0.61818

Chapter 5 **Exercise 66** Page 180

PROBLEM STATEMENT

Chapter 5, Page 180

Step 1

Total number of companies = 20

Number of CEOs who made more than $1 million = 6 \ a) P(CEO made more than $1 million) =$6/20 = 0.3$\ b) P(CEO made more than $1 million or the shareholders lost money) = 1 - P(CEO made less than $1 million and shareholders made money) =$1 - (11/20) = 0.45$\ c) P(CEO made more than $1 million / shareholders lost money) =$$\dfrac{P(CEO\ made\ more\ than\ $1\ million\ and\ shareholders\ lost\ money)}{P(shareholders\ lost\ money)}$$ = $\frac{4/20}{7/20}$ = 4/7 = 0.5714

$d) P(Both CEOs made more than \$1 million) = \frac{\binom{6}{2}}{\binom{20}{2}} = \frac{6 * 5}{20 * 19}$ = 0.07895$

Chapter 5 **Exercise 67** Page 180

PROBLEM STATEMENT

Chapter 5, Page 180

Step 1

P(Potential market sees the advertisement) = 0.6

P(Potential market remembers the advertisement / they read the advertisement) = 0.85

a) P(Potential market sees and remembers the advertisement) = P(Potential market remembers the advertisement / they read the advertisement) * P(they read the advertisement) =$0.85 * 0.6 = 0.51$

b) P(Potential market sees and does not remember the advertisement) = P(Potential market does not remember the advertisement / they read the advertisement) * P(they read the advertisement) = $(1 - 0.85) * 0.60 = 0.09$

Result

a) 0.51, b) 0.09

Chapter 5 **Exercise 68** Page 180

PROBLEM STATEMENT

Chapter 5, Page 180

Step 1

P(a particular vice president gets invited) = $1/4$

P(one of the four vice presidents does not get invited to attend any of the last five games) = P(in the last five games, any of the remaining 3 VPs get invited) = $(\frac{3}{4})^5 = 0.2373$

Result

0.2373

PROBLEM STATEMENT

Chapter 5, Page 180

Step 1

P(a CD is perfect) = $857/1000 = 0.857$

P(a CD is usable but has bad sectors) = $112/1000 = 0.112$

P(a CD is unusable) = $31/1000 = 0.031$

a) P(a randomly selected CD is not perfect) = 1 - P(a randomly selected CD is perfect) = 1 - 0.857 = 0.143

a) P(a CD is unusable / it is not perfect) = $\frac{P(a\ CD\ is\ unusable\ and\ not\ perfect)}{P(a\ CD\ is\ not\ perfect)} = \frac{0.031}{0.112 + 0.031} = 0.21678$

Result

a) 0.143, b) 0.21678

PROBLEM STATEMENT

Chapter 5, Page 180

Step 1

$P(\text{bank stock will appreciate over a year}) = 0.70$ and
$P(\text{electric utility will appreciate over a year}) = 0.60.$

a.

$$\begin{aligned} P(\text{both stocks appreciate during during the period}) =& P(\text{bank stock will appreciate over a year}) \times \\ & P(\text{electric utility will appreciate over a year}) \\ =& 0.7 \times 0.6 \\ =& 0.42 \end{aligned}$$

b.

$$\begin{aligned} P(\text{the bank stock appreciates but the utility does not}) &= P(\text{bank stock will appreciate}) \\ &\times P(\text{electric utility will not appreciate}) \\ &= 0.7 \times (1 - 0.6) \\ &= 0.28 \end{aligned}$$

c.

$$\begin{aligned} P(\text{at least one of the stocks appreciates}) &= 1 - P(\text{none of the stocks will appreciate}) \\ &= (1 - 0.7) \times (1 - 0.6) \\ &= 0.12 \end{aligned}$$

Result

a. 0.42

b. 0.28

c. 0.12

PROBLEM STATEMENT

Chapter 5, Page 180

Step 1

Let A_1 be the event that the prospects are good, A_2 be the event that the prospects are fair and A_3 be the event that the prospects are poor. Let B be the event that the profits will be made.

$P(A_1) = 0.6$, $P(A_2) = 0.3$, $P(A_3) = 0.1$, $P(B|A_1) = 0.8$, $P(B|A_2) = 0.6$, $P(B|A_3) = 0.2$. Now,

$$\begin{aligned} P(A_3|B) &= \frac{P(A_3)P(B|A_3)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3)} \\ &= \frac{0.1 \times 0.2}{0.6 \times 0.8 + 0.3 \times 0.6 + 0.1 \times 0.2} \\ &= \frac{0.02}{0.48 + 0.18 + 0.02} \\ &= \frac{0.02}{0.68} \\ &= 0.02941 \end{aligned}$$

Result

$$0.02941$$

PROBLEM STATEMENT

Chapter 5, Page 180

Step 1

DEFINITIONS

General multiplication rule:

$$P(A \text{ and } B) = P(A \cap B) = P(A) \times P(B|A) = P(B) \times P(A|B)$$

Addition rule for disjoint or mutually exclusive events:

$$P(A \cup B) = P(A \text{ or } B) = P(A) + P(B)$$

Definition **Conditional probability**:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A \text{ and } B)}{P(A)}$$

Step 2

DEFINITIONS

Let M=mesh polo shirt, S=Super T-shirt, B=Box 1 and $\sim B$=Box 2.

One of the boxes was selected at random and thus we have 1 chance in 2 of selecting each box.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(B) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{1}{2} = 0.5$$
$$P(\sim B) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{1}{2} = 0.5$$

25 of the $25 + 15 = 40$ shirts in Box 1 are mesh polo shirts, while 30 of the $30 + 10 = 40$ shirts in Box 2 are mesh polo shirts.

$$P(M|B) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{25}{40} = 0.625$$
$$P(M|\sim B) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{30}{40} = 0.75$$

Use the general multiplication rule:

$$P(B \text{ and } M) = P(B) \times P(M|B) = 0.5 \times 0.625 = 0.3125$$
$$P(\sim B \text{ and } M) = P(\sim B) \times P(M|\sim B) = 0.5 \times 0.75 = 0.375$$

Use the addition rule for mutually exclusive events (as the selected shirt cannot originate from both boxes).

$$\begin{aligned} P(M) &= P(B \text{ and } M) + P(\sim B \text{ and } M) \\ &= 0.3125 + 0.375 \\ &= 0.6875 \end{aligned}$$

Step 3

Use the definition of conditional probability:

$$\begin{aligned} P(B|M) &= \frac{P(B \text{ and } M)}{P(M)} \\ &= \frac{0.3125}{0.6875} \\ &= \frac{3125}{6875} \\ &= \frac{5}{11} \\ &\approx 0.4545 \\ &= 45.45\% \end{aligned}$$

Result

$$\frac{5}{11} \approx 0.4545 = 45.45\%$$

PROBLEM STATEMENT

Chapter 5, Page 181

Step 1

Let A be the event that the customer gets a large pizza and B be the event that the customer receives ice-cream.

$P(A) = \frac{1}{10}$ and $P(B) = \frac{1}{50}$.

a.

$$\begin{aligned} P(A \text{ or } B) &= P(A) + P(B) \\ &= \frac{1}{10} + \frac{1}{50} \\ &= \frac{6}{50} \\ &= \frac{3}{25} \end{aligned}$$

b. The customer will not win a prize if he neither gets A nor B.

$$\begin{aligned} P(\text{neither } A \text{ nor } B) &= 1 - P(A \text{ or } B) \\ &= 1 - \frac{3}{25} \\ &= \frac{22}{25} \end{aligned}$$

c. Probability that you will not win prize on three consecutive visits is

$$\begin{aligned} &= \left(\frac{22}{25}\right)^3 \\ &= 0.6815 \end{aligned}$$

d. Probability that the customer wins at least one prize on one of your next three visits is

$$\begin{aligned} &= 1 - 0.6815 \\ &= 0.3185 \end{aligned}$$

🎯 Result

a. $\frac{3}{25}$

b. $\frac{22}{25}$

c. 0.6815

d. 0.3185

PROBLEM STATEMENT

Chapter 5, Page 181

Step 1

a. First digit of the number can take any digit between 1 to 9, so first digit can be filled in 9 ways. The second digit can be filled by any number between 0 to 9 except the number that has been put as the first digit. So, second number can be filled in 9 ways. Similarly, third digit can be filled in 8 ways.

Hence, total possible three digit numbers are $= 9 \times 9 \times 8 = 648$

b. You can win the game only if your number matches the number announced on TV and that can happen in only one way. So,

$$\begin{aligned} P(\text{you win the game}) &= \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}} \\ &= \frac{1}{648} \\ &= 0.00154 \end{aligned}$$

c. Probability that you will not win with any of the tickets is

$$\begin{aligned} &= \text{Probability that you don not win on first ticket} \times \text{Probability that you don not win on second tic} \\ &\times \text{Probability that you don not win on third ticket} \\ &= (1 - 0.00154) \times (1 - 0.00154) \times (1 - 0.00154) \\ &= 0.9954 \end{aligned}$$

Result

a. 648

b. 0.00154

c. 0.9954

PROBLEM STATEMENT

Chapter 5, Page 181

Step 1

Fundamental counting principle (multiplication rule): If the first event could occur in m ways and the second event could occur in n ways, then the number of ways that the two events could occur in sequence is $m \cdot n$.

There are 2 options for each of the toppings: include in the hamburger or not include in the hamburger.

Mustard: 2 ways

Ketchup: 2 ways

Onion: 2 ways

Pickle: 2 ways

Tomato: 2 ways

Relish: 2 ways

Mayonnaise: 2 ways

Lettuce: 2 ways

Use the fundamental counting principle:

$$2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 2^8 = 256$$

Thus there are 256 different ways to order a hamburger and thus the advertisement is correct.

🎯 Result

The advertisement is correct.

PROBLEM STATEMENT

Chapter 5, Page 181

Step 1

Let A be the event that the tourists visited Beijing and B be the event that tourists visited Xi'an.

$P(A) = 0.6$, $P(B) = 0.4$ and $P(A \text{ and } B) = 0.3$.

Probability that the tourists visited at least one of the three places is that he either visits A or B. So,

$$\begin{aligned} P(A \text{ or } B) &= P(A) + P(B) - P(A \text{ and } B) \\ &= 0.6 + 0.4 - 0.3 \\ &= 0.7 \end{aligned}$$

Result

$$0.7$$

PROBLEM STATEMENT

Chapter 5, Page 181

Step 1

Probability that a person chewing the gum is successful in stopping smoking= 0.6.

Probability that in a group of 4 smokers, at least one quits smoking is

$$\begin{aligned} &= 1 - \text{Probability that no one quits smoking} \\ &= (1-0.6)^4 \\ &= 0.0256 \end{aligned}$$

Result

$$0.0256$$

PROBLEM STATEMENT

Chapter 5, Page 181

Step 1

Fundamental counting principle (multiplication rule): If the first event could occur in m ways and the second event could occur in n ways, then the number of ways that the two events could occur in sequence is $m \cdot n$.

There are 5 possible exterior designs that can be combined with any of the 3 possible interior plans.

Exterior: 5 ways

Interior: 3 ways

Use the fundamental counting principle:

$$5 \cdot 3 = 15$$

Thus there are 15 different ways that the exterior and interior plants can be offered to potential home buyers.

🎯 Result

15 different ways

PROBLEM STATEMENT

Chapter 5, Page 181

Step 1

Let A be the event that the car has a defective brakes and B be the event that the car has a defective steering.

$P(A) = 0.15$ and $P(B) = 0.05$. Since A and B are independent, $P(A \text{ and } B) = P(A)P(B)$.

a. Probability that a car is a lemon i.e has one of the problems is

$$\begin{aligned} P(A \text{ or } B) &= P(A) + P(B) - P(A \text{ and } B) \\ &= P(A) + P(B) - P(A)P(B) \\ &= 0.15 + 0.05 - 0.15 \times 0.05 \\ &= 0.2 - 0.0075 \\ &= 0.1925 \end{aligned}$$

b. Probability that a car is a hazard i.e has both problems is

$$\begin{aligned} P(A \text{ and } B) &= P(A)P(B) \\ &= 0.15 \times 0.05 \\ &= 0.0075 \end{aligned}$$

Result

a. 0.1925

b. 0.0075

PROBLEM STATEMENT

Chapter 5, Page 181

Step 1

The three numbers can be filled by digits between 0 and 9. So, the three numbers can be arranged in $10 \times 10 \times 10 = 1000$. The three letters can be filled with alphabets from A to Z, The number of ways three letters can be arranged is $26 \times 26 \times 26 = 17576$. So, total number of ways, number plate can be arranged is $17576 \times 1000 = 17576000$.

🎯 Result

$$17576000$$

PROBLEM STATEMENT

Chapter 5, Page 181

Step 1

Let M be the event that the applicant is a male and F be the event that the applicant is female. Let B be the event that the applicant is above 60 years old.

$P(M) = \frac{2}{4} = 0.5$, $P(F) = \frac{2}{4} = 0.5$, $P(B) = \frac{3}{4} = 0.75$ and $P(B|F) = 0.5$

a. $P(F \text{ and } B)$

$$\begin{aligned} P(F \text{ and } B) &= P(B|F)P(F) \\ &= 0.5 \times 0.5 \\ &= 0.25 \end{aligned}$$

b. $P(\sim B|M)$

$$\begin{aligned} P(\sim B|M) &= \frac{P(\sim B \text{ and } M)}{P(M)} \\ &= \frac{0}{0.5} \end{aligned}$$

since it is given that 1 out of 3 applicants is above 60, the rest 2 applicants are male

So, $P(\sim B|M) = 0$.

c. $P(F|B)$

$$\begin{aligned} P(F|B) &= \frac{P(F \text{ and } B)}{P(B)} \\ &= \frac{\frac{1}{4}}{0.75} \\ &= \frac{0.25}{0.75} \\ &= \frac{1}{3} \\ &= 0.33 \end{aligned}$$

🎯 Result

a. 0.25

b. 0

c. 0.33

PROBLEM STATEMENT

Chapter 5, Page 181

Step 1

DEFINITIONS

General multiplication rule:

$$P(A \text{ and } B) = P(A \cap B) = P(A) \times P(B|A) = P(B) \times P(A|B)$$

Complement rule:

$$P(\sim A) = P(\text{not } A) = 1 - P(A)$$

Step 2

SOLUTION

(a) 6 of the $4 + 6 = 10$ purchased tracts of land are in Newburg Woords.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{First Newburg}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{6}{10} = \frac{3}{5}$$

Once a tract of land in Newburg Woods has been sold, there are 9 tracts of land remaining of which 5 are in Newburg Woods.

$$P(\text{Second Newburg}|\text{First Newburg}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{5}{9}$$

Use the general multiplication rule:

$$\begin{aligned}
&P(\text{Both Newburg}) \\
&= P(\text{First Newburg}) \\
&\times P(\text{Second Newburg}|\text{First Newburg}) \\
&= \frac{6}{10} \times \frac{5}{9} \\
&= \frac{3}{5} \times \frac{5}{9} \\
&= \frac{3 \times 5}{5 \times 9} \\
&= \frac{3}{9} \\
&= \frac{1}{3} \\
&\approx 0.3333 \\
&= 33.33\%
\end{aligned}$$

Step 3

(b) 6 of the $4 + 6 = 10$ purchased tracts of land are in Newburg Woords.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{First Newburg}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{6}{10} = \frac{3}{5}$$

Once a tract of land in Newburg Woods has been sold, there are 9 tracts of land remaining of which 5 are in Newburg Woods.

$$P(\text{Second Newburg}|\text{First Newburg}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{5}{9}$$

Once two tracts of land in Newburg Woods have been sold, there are 8 tracts of land remaining of which 4 are in Newburg Woods.

$$P(\text{Third Newburg}|\text{First two Newburg}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{4}{8}$$

Once three tracts of land in Newburg Woods have been sold, there are 7 tracts of land remaining of which 3 are in Newburg Woods.

$$P(\text{Fourth Newburg}|\text{First three Newburg}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{3}{7}$$

Use the general multiplication rule:

$$\begin{aligned}
&P(\text{All 4 Newburg woods}) \\
&= P(\text{First Newburg}) \\
&\times P(\text{Second Newburg }|\text{First Newburg}) \\
&\times P(\text{Third Newburg }|\text{First two Newburg}) \\
&\times P(\text{Fourth Newburg }|\text{First three Newburg}) \\
&= \frac{6}{10} \times \frac{5}{9} \times \frac{4}{8} \times \frac{3}{7} \\
&= \frac{3}{5} \times \frac{5}{9} \times \frac{1}{2} \times \frac{3}{7} \\
&= \frac{3 \times 5 \times 1 \times 3}{5 \times 9 \times 2 \times 7} \\
&= \frac{45}{630} \\
&= \frac{1}{14}
\end{aligned}$$

Step 4

Use the complement rule:

$$\begin{aligned} P(\text{At least 1 Holly Farms}) &= 1 - P(\text{All 4 Newburg woods}) \\ &= 1 - \frac{1}{14} \\ &= \frac{13}{14} \\ &\approx 0.9286 \\ &= 92.86\% \end{aligned}$$

Step 5

(c) Two events are **independent**, if the probability that one event occurs in no way affects the probability of the other event occurring.

The events are not independent (dependent) in this case, because you cannot sell a tract of land if you already sold it on one of the previous sales and thus the probability of selling a particular tract of land is affected by those that were previously sold.

🎯 Result

(a) $\frac{1}{3} \approx 0.3333 = 33.33\%$

(b) $\frac{13}{14} \approx 0.9286 = 92.86\%$

(c) Dependent

PROBLEM STATEMENT

Chapter 5, Page 181

Step 1

Each of the four blanks of the password can be filled in 26 ways, so the different number of passwords possible are $26 \times 26 \times 26 = 17576$

Result

$$17576$$

PROBLEM STATEMENT

Chapter 5, Page 181

Step 1

Probability that a can is contaminated is $=\frac{1}{24}$

a. Total combinations of 3 cans is ${}^{24}C_3 = \frac{24!}{3!(24-3)!} = \frac{24 \times 23 \times 22}{3 \times 2} = 2024.$

b. Probability that the contaminated can is selected for testing is

$$\begin{aligned} &= \frac{{}^{24}C_1}{2024} \\ &= \frac{24}{2024} \\ &= 0.01186 \end{aligned}$$

Result

a. 12144

b. 0.01186

PROBLEM STATEMENT

Chapter 5, Page 182

Step 1

Fundamental counting principle: If the first event could occur in m ways and the second event could occur in n ways, then the number of ways that the two events could occur in sequence is $m \cdot n$.

There are 10 possible U.S. presidents and 10 possible vice presidents (as we have the vice president corresponding to each U.S. president).

U.S. President: 10 ways

Vice President: 10 ways

Use the fundamental counting principle:

$$10 \cdot 10 = 10^2 = 100$$

Thus there are 100 possible matches when making the matches randomly.

Step 2

1 of these 100 possible matches will consist of all 10 correct matches between U.S. president and vice president.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{all correct}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{1}{100} = 0.01 = 1\%$$

🎯 Result

100 possible matches

$$\frac{1}{100} = 0.01 = 1\%$$

PROBLEM STATEMENT

Chapter 5, Page 182

Step 1

Probability that A works is $P(A) = 0.9$ and the probability that B works is $P(B) = 0.9$.

Since A and B are independent, $P(A \text{ and } B = P(A)P(B)$.

Probability that the system operate is

$$\begin{aligned} &= P(A \text{ and } B) \\ &= P(A)P(B) \\ &= 0.9 \times 0.9 \\ &= 0.81 \end{aligned}$$

Result

$$0.81$$

PROBLEM STATEMENT

Chapter 5, Page 182

Step 1

Let A_1 be the event of Tyson supplying the tubes, A_2 be the event of Fuji supplying the tubes, A_3 be the event of Kirkpatricks supplying the tubes and A_4 be the event of Parts Inc supplying the tubes. Let B be the event of getting the defective $P(A_1) = 0.2$, $P(A_2) = 0.3$, $P(A_3) = 0.25$, $P(A_4) = 0.25$, $P(B|A_1) = 0.03$, $P(B|A_2) = 0.04$ and $P(B|A_3) = 0.07$ and $P(B|A_4) = 0.065$.

a. Overall percent defective

$$\begin{aligned}
&= P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3) + P(A_4)P(B|A_4) \\
&= 0.2 \times 0.03 + 0.3 \times 0.04 + 0.25 \times 0.07 + 0.25 \times 0.065 \\
&= 0.006 + 0.012 + 0.0175 + 0.01625 \\
&= 0.05175
\end{aligned}$$

b. Probability that the defective tube is from Tyson

$$\begin{aligned}
P(A_1|B) &= \frac{P(A_1)P(B|A_1)}{P(A_1)P(B|A_1) + P(A_2)P(B|A_2) + P(A_3)P(B|A_3) + P(A_4)P(B|A_4)} \\
&= \frac{0.2 \times 0.03}{0.2 \times 0.03 + 0.3 \times 0.04 + 0.25 \times 0.07 + 0.25 \times 0.065} \\
&= \frac{0.006}{0.006 + 0.012 + 0.0175 + 0.01625} \\
&= \frac{0.006}{0.05175} \\
&= 0.116
\end{aligned}$$

Result

a. 0.05175

b. 0.116

PROBLEM STATEMENT

Chapter 5, Page 182

Step 1

Let G be the event that driver is good, M be the event that the driver is medium and P be the event that the driver is poor. Let B be the event of an accident.

$P(G) = 0.3$, $P(M) = 0.5$ and $P(B) = 0.2$. $P(B|G) = 0.01$, $P(B|M) = 0.03$ and $P(B|P) = 0.1$.

a. Probability that the driver is good

$$\begin{aligned} P(G|B) &= \frac{P(G)P(B|G)}{P(G)P(B|G) + P(M)P(B|M) + P(P)P(B|P)} \\ &= \frac{0.3 \times 0.01}{0.3 \times 0.01 + 0.5 \times 0.03 + 0.2 \times 0.1} \\ &= \frac{0.003}{0.003 + 0.015 + 0.02} \\ &= \frac{0.003}{0.038} \\ &= 0.0789 \end{aligned}$$

b. Probability that the driver is medium

$$\begin{aligned} P(M|B) &= \frac{P(M)P(B|M)}{P(G)P(B|G) + P(M)P(B|M) + P(P)P(B|P)} \\ &= \frac{0.5 \times 0.03}{0.3 \times 0.01 + 0.5 \times 0.03 + 0.2 \times 0.1} \\ &= \frac{0.015}{0.003 + 0.015 + 0.02} \\ &= \frac{0.015}{0.038} \\ &= 0.395 \end{aligned}$$

Step 2

c. Probability that the driver is poor

$$
\begin{aligned}
P(P|B) &= \frac{P(P)P(B|P)}{P(G)P(B|G) + P(M)P(B|M) + P(P)P(B|P)} \\
&= \frac{0.2 \times 0.1}{0.3 \times 0.01 + 0.5 \times 0.03 + 0.2 \times 0.1} \\
&= \frac{0.02}{0.003 + 0.015 + 0.02} \\
&= \frac{0.02}{0.038} \\
&= 0.395
\end{aligned}
$$

🎯 Result

a. 0.0789

b. 0.395

c. 0.526

PROBLEM STATEMENT

Chapter 5, Page 182

Step 1

Probability that a flight is on time is 0.8.

Probability that all flights arrive on time is

$$\begin{aligned} &= 0.8 \times 0.8 \times 0.8 \ \ (\text{since flights are independent}) \\ &= 0.512 \end{aligned}$$

🎯 Result

$$0.512$$

PROBLEM STATEMENT

Chapter 5, Page 182

Step 1

Probability of HP server being down is 0.05.

Probability that at least one of them is operational is

$$
\begin{aligned}
&= 1 - \text{Probability that none of them is operational} \\
&= 1 - (1 - 0.05)^3 \\
&= 1 - (0.95)^3 \\
&= 1 - 0.857 \\
&= 0.143
\end{aligned}
$$

Result

$$0.143$$

PROBLEM STATEMENT

Chapter 5, Page 182

Step 1

Probability that an LCD is manufactured by Samsung is 0.22.

Probability that at least one of the three LCDs is samsung is

$$\begin{aligned} &= 1 - \text{none of the LCD is samsung} \\ &= 1 - (1 - 0.22)^3 \\ &= 1 - (0.78)^3 \\ &= 0.5254 \end{aligned}$$

Result

$$0.5254$$

PROBLEM STATEMENT

Chapter 5, Page 182

Step 1

DEFINITIONS

General addition rule for any two events:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Definition **Conditional probability**:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A \text{ and } B)}{P(A)}$$

Step 2

SOLUTION

(a) (1) 67 of the 105 homes have a pool.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{pool}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{67}{105}$$

15 of the 105 homes are in township 1.

$$P(\text{township 1}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{15}{105}$$

6 of the 105 homes are in township 1 and have a pool.

$$P(\text{township 1 and pool}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{6}{105}$$

Use the general addition rule:

$$\begin{aligned}
&P(\text{township 1 or pool}) \\
&= P(\text{township 1}) + P(\text{pool}) - P(\text{township 1 and pool}) \\
&= \frac{67}{105} + \frac{15}{105} - \frac{6}{105} \\
&= \frac{67 + 15 - 6}{105} \\
&= \frac{76}{105} \\
&\approx 0.7238 \\
&= 72.38\%
\end{aligned}$$

Step 3

(2) 25 of the 105 homes are in township 3.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{township 3}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{25}{105}$$

18 of the 105 homes are in township 3 and have a pool.

$$P(\text{township 3 and pool}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{18}{105}$$

Use the definition of conditional probability

$$\begin{aligned} &P(\text{pool}|\text{township 3}) \\ &= \frac{P(\text{township 3 and pool})}{P(\text{township 3})} \\ &= \frac{18/105}{25/105} \\ &= \frac{18}{25} \\ &= 0.72 \\ &= 72\% \end{aligned}$$

Step 4

(3) 18 of the 105 homes are in township 3 and have a pool.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$\begin{aligned} P(\text{township 3 and pool}) &= \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} \\ &= \frac{18}{105} \\ &= \frac{6}{35} \\ &\approx 0.1714 \\ &= 17.14\% \end{aligned}$$

Step 5

(b) (1) 71 of the 105 homes have a garage attached.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$\begin{aligned} P(\text{garage}) &= \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} \\ &= \frac{71}{105} \\ &\approx 0.6762 \\ &= 67.62\% \end{aligned}$$

Step 6

(2) 16 of the 105 homes are in township 5.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{township 5}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{16}{105}$$

4 of the 105 homes are in township 5 and do not have a garage.

$$P(\text{township 5 and no garage}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{4}{105}$$

Use the definition of conditional probability

$$\begin{aligned} &P(\text{no garage}|\text{township 5}) \\ &= \frac{P(\text{township 5 and no garage})}{P(\text{township 5})} \\ &= \frac{4/105}{16/105} \\ &= \frac{4}{16} \\ &= \frac{1}{4} \\ &= 0.25 \\ &= 25\% \end{aligned}$$

Step 7

(3) 15 of the 105 homes are in township 3 and have a garage attached.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$\begin{aligned} P(\text{township 3 and garage}) &= \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} \\ &= \frac{15}{105} \\ &= \frac{1}{7} \\ &\approx 0.1429 \\ &= 14.29\% \end{aligned}$$

Step 8

(4) 67 of the 105 homes have no garage.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{no garage}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{67}{105}$$

20 of the 105 homes are in township 2.

$$P(\text{township 2}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{20}{105}$$

34 of the 105 homes are in township 2 and have no garage.

$$P(\text{township 2 and no garage}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{34}{105}$$

Use the general addition rule:

$$\begin{aligned}
&P(\text{township 2 or no garage}) \\
&= P(\text{township 2}) + P(\text{no garage}) - P(\text{township 2 and no garage}) \\
&= \frac{20}{105} + \frac{34}{105} - \frac{15}{105} \\
&= \frac{20 + 34 - 15}{105} \\
&= \frac{39}{105} \\
&= \frac{13}{35} \\
&\approx 0.3714 \\
&= 37.14\%
\end{aligned}$$

🎯 Result

(a)

(1) $\frac{76}{105} \approx 0.7238 = 72.38\%$

(2) $\frac{18}{25} \approx 0.72 = 72\%$

(3) $\frac{6}{35} \approx 0.1714 = 17.14\%$

(b)

(1) $\frac{71}{105} \approx 0.6762 = 67.62\%$

(2) $\frac{1}{4} = 0.25 = 25\%$

(3) $\frac{1}{7} \approx 0.1429 = 14.29\%$

(4) $\frac{13}{35} \approx 0.3714 = 37.14\%$

PROBLEM STATEMENT

Chapter 5, Page 182

Step 1

DEFINITIONS

General addition rule for any two events:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Definition **Conditional probability**:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A \text{ and } B)}{P(A)}$$

Step 2

SOLUTION

(a) A team is considered to have a winning season when it won at least 81 games. All other teams have a losing season.

We divide the attendance into three categories: less than 2.0 million, 2.0 to 3.0 million, more than 3.0 million.

We count the number of teams that fall within each combination of categories and summarize the results in a table:

	<2.0 million	2.0 to 3.0 million	>3.0 million	Total
Winning Season	2	7	7	16
Losing Season	9	3	2	14
Total	11	10	9	30

(1) 16 of the 30 teams had a winning season.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{Winning}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{16}{30} = \frac{8}{15} \approx 0.5333 = 53.33\%$$

Step 3

(2) 16 of the 30 teams had a winning season.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{Winning}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{16}{30}$$

9 of the 30 teams had an attendance of more than 3.0 million.

$$P(>\ 3.0\text{ million}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{9}{30}$$

7 of the 30 teams had a winning season and an attendance of more then 3.0 million.

$$P(\text{Winning and }>\ 3.0\text{ million}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{7}{30}$$

Use the general addition rule:

$$\begin{aligned}
&P(\text{Winning or }>\ 3.0\text{ million}) \\
&= P(\text{Winning}) + P(>\ 3.0\text{ million}) \\
&- P(\text{Winning and }>\ 3.0\text{ million}) \\
&= \frac{16}{30} + \frac{9}{30} - \frac{7}{30} \\
&= \frac{16+9-7}{30} \\
&= \frac{18}{30} \\
&= \frac{3}{5} \\
&= 0.6 \\
&= 60\%
\end{aligned}$$

Step 4

(3) 9 of the 30 teams had an attendance of more than 3.0 million.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(>\ 3.0 \text{ million}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{9}{30}$$

7 of the 30 teams had a winning season and an attendance of more then 3.0 million.

$$P(\text{Winning and } >\ 3.0 \text{ million}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{7}{30}$$

Use the definition of conditional probability

$$\begin{aligned} &P(\text{Winning}|>\ 3.0 \text{ million}) \\ &= \frac{P(\text{Winning and } >\ 3.0 \text{ million})}{P(>\ 3.0 \text{ million})} \\ &= \frac{7/30}{9/30} \\ &= \frac{7}{9} \\ &\approx 0.7778 \\ &= 77.78\% \end{aligned}$$

Step 5

(4) 2 of the 30 teams had a winning season and an attendance of less than 2.0 million.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$\begin{aligned} P(\text{Winning and } <\ 2.0 \text{ million}) &= \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} \\ &= \frac{2}{30} \\ &= \frac{1}{15} \\ &\approx 0.0667 \\ &= 6.67\% \end{aligned}$$

Step 6

(b) A team is considered to have a winning season when it won at least 81 games. All other teams have a losing season.

We divide the year that the stadium was built into three categories: new (built in $2009 - 15 = 1994$ or later) and old (built before 1994). Note that we assume that the year is 2009 and that a stadium is new when it was built in the 15 years prior to 2009.

We count the number of teams that fall within each combination of categories and summarize the results in a table:

	Winning	Losing	Total
New	8	8	16
Old	8	6	14
Total	16	14	30

(1) 16 of the 30 teams had a winning season.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{Winning}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{16}{30} = \frac{8}{15} \approx 0.5333 = 53.33\%$$

Step 7

(2) 8 of the 30 teams had a winning season and played in a new stadium.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$\begin{aligned} P(\text{Winning and new}) &= \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} \\ &= \frac{8}{30} \\ &= \frac{4}{15} \\ &\approx 0.2667 \\ &= 26.67\% \end{aligned}$$

Step 8

(3) 16 of the 30 teams had a winning season.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{Winning}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{16}{30}$$

16 of the 30 teams played in a new stadium.

$$P(\text{township 1}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{16}{30}$$

8 of the 30 teams had a winning season and played in a new stadium.

$$P(\text{Winning and new}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{8}{30}$$

Use the general addition rule:

$$\begin{aligned}
&P(\text{township 1 or pool}) \\
&= P(\text{township 1}) + P(\text{pool}) - P(\text{township 1 and pool}) \\
&= \frac{16}{30} + \frac{16}{30} - \frac{8}{30} \\
&= \frac{16 + 16 - 8}{30} \\
&= \frac{24}{30} \\
&= \frac{4}{5} \\
&= 0.8 \\
&= 80\%
\end{aligned}$$

🎯 Result

(a)

(1) $\frac{8}{15} \approx 0.5333 = 53.33\%$

(2) $\frac{3}{5} = 0.6 = 60\%$

(3) $\frac{7}{9} \approx 0.7778 = 77.78\%$

(4) $\frac{1}{15} \approx 0.0667 = 6.67\%$

(b)

(1) $\frac{8}{15} \approx 0.5333 = 53.33\%$

(2) $\frac{4}{15} \approx 0.2667 = 26.67\%$

(3) $\frac{4}{5} = 0.8 = 80\%$

PROBLEM STATEMENT

Chapter 5, Page 183

Step 1

DEFINITIONS

General addition rule for any two events:

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

Definition **Conditional probability**:

$$P(B|A) = \frac{P(A \cap B)}{P(A)} = \frac{P(A \text{ and } B)}{P(A)}$$

Step 2

SOLUTION

Order the data values from smallest to largest:

329, 329, 337, 355, 357, 359, 369, 380, 381, 382, 390, 390, 392, 392, 396, 403, 406, 410, 411, 414, 422, 423, 424, 426, 427, 428, 432, 432, 433, 433, 436, 439, 441, 442, 444, 448, 449, 450, 452, 455, 457, 458, 459, 459, 461, 462, 466, 467, 468, 469, 469, 471, 474, 474, 475, 476, 477, 478, 478, 489, 492, 493, 493, 494, 496, 497, 501, 503, 503, 504, 504, 505, 514, 515, 529, 540, 546, 558, 561, 570

The **median** is the middle value of the sorted data set. Since there are 80 data values in this case, the median is the average of the 40th and 41st data value.

$$m = \frac{455 + 457}{2} = \frac{912}{2} = 456$$

Thus the median maintenance cost is $\$456$.

Next, we determine a contingency table that contains the rows new (less than 5 years old), medium (at least 5 but less than 10 years), and old (10 or more years), while it contains the columns' maintenance cost below $\$456$ and maintenance cost above $\$456$.

We count the number of data values that corresponds with each combination of age and maintenance cost.

	Maintenance cost <$456	Maintenance cost >$456	Total
New	13	5	18
Medium	26	21	47
Old	1	14	15
Total	40	40	80

Step 3

(a) 18 of the 80 buses are new.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{new}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{18}{80} = \frac{9}{40} = 0.225 = 22.5\%$$

Thus 22.5% of the buses are new.

Step 4

(b) 18 of the 80 buses are new.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{new}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{18}{80}$$

13 of the 80 buses are new and have a low maintenance.

$$P(\text{new and low}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{13}{80}$$

Use definition conditional probability:

$$\begin{aligned} P(\text{low}|\text{new}) &= \frac{P(\text{new and low})}{P(\text{new})} \\ &= \frac{13/80}{18/80} \\ &= \frac{13}{18} \\ &\approx 0.7222 \\ &= 72.22\% \end{aligned}$$

Thus 72.22% of the new buses have low maintenance.

Step 5

(c) 15 of the 80 buses are old.

The probability is the number of favorable outcomes divided by the number of possible outcomes:

$$P(\text{old}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{15}{80}$$

14 of the 80 buses are old and have a high maintenance.

$$P(\text{old and high}) = \frac{\#\text{ of favorable outcomes}}{\#\text{ of possible outcomes}} = \frac{14}{80}$$

Use definition conditional probability:

$$\begin{aligned} P(\text{high}|\text{old}) &= \frac{P(\text{old and high})}{P(\text{old})} \\ &= \frac{14/80}{15/80} \\ &= \frac{14}{15} \\ &\approx 0.9333 \\ &= 93.33\% \end{aligned}$$

Thus 93.33% of the old buses have high maintenance.

Step 6

(d) By part (b), we know that most new buses have low maintenance.

By part (c), we know that most old buses require high maintenance.

This then implies that the maintenance cost seems to be related to the age of the bus, because the maintenance cost appears to increase as the age increases.

🎯 Result

(a) 22.5%

(b) 72.22%

(c) 93.33%

(d) Related